

Assignment: Arrays in C Programming

1. 1D Array: dailySteps

```
#include <stdio.h>

int main()
{
    int dailySteps[7] = {5000, 6200, 7100, 8000, 6500, 9000, 7500};

    printf("Daily Steps for the Week:\n");

    for(int i = 0; i < 7; i++)
    {
        printf("Day %d: %d steps\n", i + 1, dailySteps[i]);
    }

    return 0;
}
```

Sample Output

Daily Steps for the Week:

Day 1: 5000 steps

Day 2: 6200 steps

Day 3: 7100 steps

Day 4: 8000 steps

Day 5: 6500 steps

Day 6: 9000 steps

Day 7: 7500 steps

2. 2D Array: playlistRatings

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int playlistRatings[3][5] =
```

```
    {
```

```
        {5, 4, 5, 4, 5},
```

```
        {3, 4, 4, 5, 5},
```

```
        {5, 5, 4, 5, 4}
```

```
    };
```

```
    printf("Ratings for the Second Playlist:\n");
```

```
    for(int day = 0; day < 5; day++)
```

```
    {
```

```
        printf("Day %d: %d\n", day + 1, playlistRatings[1][day]);
```

```
    }
```

```
    return 0;
```

```
}
```

Sample Output

Ratings for the Second Playlist:

Day 1: 3

Day 2: 4

Day 3: 4

Day 4: 5

Day 5: 5

3. Function to Calculate Average Weekly Zomato Spending

```
#include <stdio.h>
```

```
float calculateAverage(int orders[], int size)
```

```
{
```

```
    int sum = 0;
```

```
    for(int i = 0; i < size; i++)
```

```
    {
```

```
        sum += orders[i];
```

```
    }
```

```
    return (float)sum / size;
```

```
}
```

```
int main()
```

```
{
```

```
    int zomatoOrders[7] = {250, 180, 320, 150, 400, 280, 220};
```

```
    float average = calculateAverage(zomatoOrders, 7);
```

```
    printf("Average Weekly Spend: ₹%.2f\n", average);
```

```
    return 0;
}
```

Sample Output

Average Weekly Spend: ₹257.14

4. Highest Score from Each IPL Match

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int cricketScores[4][2] =
```

```
    {
```

```
        {180, 175},
```

```
        {210, 198},
```

```
        {165, 172},
```

```
        {225, 220}
```

```
    };
```

```
    printf("Highest Score from Each Match:\n");
```

```
    for(int match = 0; match < 4; match++)
```

```
    {
```

```
        int highest;
```

```
        if(cricketScores[match][0] > cricketScores[match][1])
```

```
{
    highest = cricketScores[match][0];
}
else
{
    highest = cricketScores[match][1];
}

printf("Match %d: %d runs\n", match + 1, highest);
}

return 0;
}
```

Sample Output

Highest Score from Each Match:

Match 1: 180 runs

Match 2: 210 runs

Match 3: 172 runs

Match 4: 225 runs

Conclusion

This assignment demonstrates the use of:

- One-Dimensional Arrays
- Two-Dimensional Arrays
- Array Traversal using Loops
- Functions with Arrays

- Average Calculation
- Finding Maximum Values in a 2D Array

These concepts are fundamental for storing and processing collections of data in C programming.